

1. Research Proposal: Anchor & Adapt: Architectural Blueprints for Managing Semantic Decoupling in Multi-Agent Knowledge Graphs

1.1. Introduction & Objectives

Automated Scientific Knowledge Graph Construction (AS-KGC) and GraphRAG deploy LLM-driven multi-agent pipelines to transform raw text into structured networks. To prevent chaotic search spaces, systems are initialized with a core seed graph (ontological schema or anchor triples). This creates a structural tension: balancing core scaffolding (**integrity**) with open-world evolution (**discovery**). When mismanaged, systems suffer from **Semantic Decoupling** — where peripheral agent expansions progressively warp away from the parent core’s logical parameters. This manifests in three bottlenecks: - **Scale-Boundary Dilemma**: Seeds that are too narrow cause cold-starts; overly broad seeds introduce relational noise.

- **Alignment Sparsity**: Cross-lingual or low-resource target domains lack dense anchor pairs, collapsing propagation into single clusters.
- **Core Mutation Shock**: New empirical data invalidates historical constraints, but updating anchors triggers catastrophic forgetting.

1.2. Research Objectives

This research introduces **Anchor & Adapt** to deliver practical blueprints that: 1. **Calibrate** initial seed “narrowness” using an automated, entropy-based boundary metric. 2. **Propagate** labels across sparse cross-lingual structures using specialized multi-agent coordination. 3. **Mutate and spawn** new sovereign core seeds dynamically via non-destructive topological splitting.

1.3. Literature Review & Research Gap

Contemporary AS-KGC leverages multi-agent orchestrations (Dong et al., 2025) to scale triple extraction, while GraphRAG paradigms (Peng et al., 2024; Zhang et al., 2025) use graph topologies to optimize reasoning. While grounding retrieval in a seed core reduces hallucinations (KG-Orchestra, 2026), **static schemas** cause severe cold-starts when encountering shifting scientific frontiers. Over time, continuous ingestion loops suffer from **relation dilution**, where precise constraints degrade into generic tokens (Iisaka, 2026). Embedding-based clustering can detect this drift in vector space (Verkijk et al., 2024), but current literature lacks remediation protocols.

1.4. Research Gap

Existing literature treats graph seeding as a binary, static choice: rigid schemas or unconstrained extraction. No framework provides systematic blueprints for a hybrid, self-evolving seed architecture that uses continuous semi-supervised tracking to balance core scaffolding with open-world evolution while actively defending against semantic decoupling.

1.5. Methodology & DSR Phases

This study follows a **Design Science Research (DSR)** approach across four primary phases:

1.5.1. Phase 1: Seed Calibration Module

To resolve seed narrowness, we compute a corpus-wide semantic entropy score (HD) from target texts. The optimal out-degree centrality (CO) and axiomatic density (Ap) for the initial core (G_{seed}) are determined via:

$$\text{Boundary Score: } \alpha(CO \cdot Ap) - \beta(HD)$$

Where α and β are tunable weights.

1.5.2. Phase 2: Cross-Lingual Agent Protocol

We deploy a decentralized agent trio:

- **Seeker:** Extracts candidate triples.
- **Librarian:** Validates and filters knowledge.
- **Alignment:** Maps cross-lingual entities.

The Alignment Agent utilizes a **Graph Neural Network (GNN)** embedding space to project sparse target-language extractions into dense source-language seeds via **iterative semi-supervised label propagation**.

1.5.3. Phase 3: Drift Detection

Every N insertions, the graph is vectorized to evaluate two drift metrics:

- **Centroid Distance Shift:** Euclidean shift between original seed centroid (C_{seed}) and new peripheral cluster centroid (C_{new}).

$$\text{Jaccard Neighborhood Drift: } J_{drift}(e) = 1 - (Nt(e) \cap Nt+1(e) / Nt(e) \cup Nt+1(e))$$

Where $Nt(e)$ is the neighborhood of entity e at time t .

1.5.4. Phase 4: Non-Destructive Mutation

When $J_{drift} \rightarrow 1$ and centroid variance thresholds (σ) are breached, a **Core Mutation Event**:

- Isolates the drifting cluster,
 - Clips weak edge bridges,
 - Indexes its center as an independent new core (G_{seed_B}),
 - Spawns new workflows branching exclusively from it.
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Evaluation: Framework performance will be benchmarked against multi-lingual PubMed/arXiv subsets, measuring **KGC Precision/Recall**, **GraphRAG Hallucination Rates**, and **structural resilience under induced paradigm shifts**.

1.6. Timeline, Feasibility & References

Months	Task
1–3	Phase 1: Seed Calibration, dataset procurement, and mathematical validation
4–6	Phase 2: Build agent protocols and deploy semi-supervised cross-lingual GNN
7–9	Phases 3 & 4: Implement Jaccard monitoring and automated core-spawning
10–12	Evaluation: GraphRAG stress tests, ablation studies, and manuscript finalization

1.7. References

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